COMP.2800: Software Development
LAB 6: Game Development Concepts in Java

LAB 5: Overview

The objective of this lab exercise is to use the Java 2D Graphics, Java Swing, and `BufferedImage` API to create a Character Customization screen.

LAB 6: Tasks

Complete the following tasks.

Task 1: `JPanel/Canvas`

Write a program with two views (i.e., `JPanels/Canvas`):

- One, displaying the character sprite, and
- The other containing Swing components (i.e., `JLabel` and `JComboBox`) for customization.

You must implement the following:

- `JComboBox` must have at least 5 options in it.
- The user can select from those options, and each option represents the specific customization that the character will use.
- The respective customization image will appear on top of the character in the other view, once the user makes a selection.

Task 2: Keyboard Input

Implement keyboard input:

- Use the WASD keys or the Arrow Keys to move the character sprite in the first view.
- Ensure that if the character is customized, the selected image moves along with it.

Task 3: Keyboard Input

Suggestions:

- You can put all the sprites, including the character image and customization images into one PNG file, or
- You can separate the sprites into two PNG files.

In either case, you must have at most 2 PNG files that you are loading.

Note: A presentation for a similar implementation can be found on the respective slide deck, published on the Lab module in Brightspace.

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LAB 6 – Submission

To formally submit your work for this Lab:

- Complete the program.
- Make sure your code compiles without errors.
- Make sure your code runs as expected.
- Make sure you have included extended comments in your code, describing the program.
- Prepare your submission:
  - Save the code file using the following naming conventions:
    - COMP.2800-Lab-x.FirstName.LastName.StudentID.java,
      replacing the underlined items with the ones relevant to your submission.
  - If you have only one Java file to submit, upload only that file.
  - If you have created multiple files:
    - Include a plain text file with explanations (such as README.MD) if there are any additional requirements to run your program.
    - Compress the project folder (zip compression) you created.
    - Rename the folder as follows: COMP.2800-Lab-x.FirstName.LastName.StudentID.zip (for example: *COMP.2800-Lab-6.Andreas.Maniatis.StudentID.zip*).
- Submit the file you created in Brightspace, in the dropbox of the respective lab.
  When submitting, please indicate in the submission textbox the Lab section you belong to.

Evaluation:

1. It is our intention to evaluate your work being done in-person, while you are in the lab.
2. For every week's Lab, it is expected that you will attend in-person, to familiarize yourselves with the Lab infrastructure, meet with the GATAs and solve problems.
3. Each session is only 80 minutes long, so it is strongly suggested that you arrive early, are focused and do not disrupt the process.
4. It is desired that you complete your work while in the lab and get graded after live review from the GA/TAs.
5. Should you need more time, you can take your work home and present it to the GA/TAs on the very next lab session (the week following).
6. You may decide to work from home, and submit your work remotely, but still, you need to have your work peer reviewed, in-person, during lab time to receive your grade.
7. Marking scheme:
   a. Load the image and draw the character (4 Marks).
   b. Move the character using the keyboard (3 Marks).
   c. Customization is working correctly (3 Marks).

   If your implementation is not correct but is on the right track (50% of the marks).

Note: Select questions from each lab may be included in your upcoming assignments (and therefore they will be required to be submitted as part of your assignment submission).

Select questions in future labs may be required to be submitted at the end of your lab session. Therefore, it is worthwhile to spend serious time to attend your lab session and completing all lab questions as much as you can.

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